GCSE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2023

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2023 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

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GCSE DIGITAL TECHNOLOGY

Chief Examiner's Report

Assessment Unit 1 Digital Technology Compulsory Core

Subject Overview

Candidates have been well prepared for the examinations and there is excellent teaching and learning occurring across the subject area. Scripts evidenced solid understanding of the key concepts underpinning Digital Technology and the evolution of skills in areas such as programming and algorithmic design.

A comprehensive range of learning outcomes from the different areas of each of the unit specifications were examined across the three papers. Questions were posed in a variety of formats to enable candidates to access marks and to display their knowledge and understanding of concepts. Candidates were also provided with significant opportunities to demonstrate their skills in areas such as spreadsheets, databases HTML and developing solutions through algorithms. The papers provided good discriminatory questions which allowed for differentiation leading to a good range of outcomes at subject level. Across all three papers candidates demonstrated a good understanding of the subject matter within the learning outcomes. This year again showed improving focus on practical programming and computational thinking evidenced by the marks achieved in questions relating to coding and algorithm.

There is still some room for improvement with regard to examination technique. Some candidates do not read carefully and consequently misinterpret the question or omit detail requested therein and are penalised when it comes to marking. Some also re-state the stem of the question or do not use the information provided, in the question, to help them in their response. Some candidates may benefit from additional guidance in this area.

Supervising Examiners Overview

Altogether the Compulsory Core paper was well planned and allowed the candidates to attain good results. The paper made provision for all levels of ability, and this was due to the formatting and variety of questions, making it very accessible to all candidates. Candidates performed well in areas such as spreadsheets, software, and the CPU (Central Processing Unit). It is clear that they are being provided with relevant practical experience to support the theoretical knowledge. The language used within the paper was appropriate for GCSE and the provided application of the mark scheme assisted the markers to award candidates accurately across all questions.

Q1 **(a)** This question was generally very well answered and was a good starting question on the paper.

(b) Part (b) was also well answered, however, some candidates did not use the correct terminology and missed out on the mark.

(c) A significant number of candidates were unable to distinguish between data and information. This is a requirement of the specification and candidates should be encouraged to learn these definitions and apply them to samples of data and information.

Q2 **(a)** This part of the question was very well answered.

(b) Some candidates confused resolution with pixels and therefore didn't always obtain a mark. Obtaining the second mark was difficult for those who got the first mark about, "increased file size."

(c) Candidates chose the correct statement about buffering.

(d) Candidates had very little difficulty with this part.

(e) Generally, the majority of candidates were able to expand the letters ADC.

(f) Again, this part was well answered by the candidates.

Q3 **(a)** This part of the question challenged numerous candidates suggesting their level of knowledge around database terminology let them down. Again, technical terms and definitions should be studied along with the practical applications.

(b) Most candidates were well informed about validation methods, and this was very pleasing, but others struggled to achieve all three.

Q4 This question was based on spreadsheets, and this is a very important skill for candidates to have especially when considering the workplace, where spreadsheets often form an important element of their workload.

(a) This part was well answered.

(b) Candidates were required to insert two simple formulas to answer this part. The mark scheme required the formulas to be written in context. However, many candidates lost marks by not being able to correctly write a spreadsheet formula.

(c) This format of question has been on the examination paper previously and it was good to acknowledge that more candidates are beginning to understand this key function.

(d) This part was well answered.

(e) Generally, well answered.

(f) Well answered.

(g) Numerous candidates did not know what a template was. This is perhaps because they were commonly used in word processing applications. Candidates should review simple template examples to ensure they have an understanding of their function.

Q5 **(a)** Most candidates could expand these three letters.

(b) Generally, well answered.

(c) Most candidates got this question correct.

(d) Some candidates did not seem to be aware of utility software and were unable to answer.

(e) The majority of candidates understood the input/output part but did not provide realistic technological examples of use.

Q6 **(a)** This part of the question was very well answered, demonstrating that candidates have a good understanding of communication media and their applications.

(b) This part of the question was poorly answered by a significant number of candidates. Candidates lost marks because they could not specify the 'physical devices'.

Q7 **(a)** Once again, most candidates correctly expanded the three letters.

(b) This part was generally well answered.

(c) This part of the question was set as a stretch and challenge and many candidates got full marks. Others got at least two. Some drew good diagrams but failed to label the diagram. Perhaps there is a need for candidates to read in more detail what is being asked.

(d) Generally, well answered.

(e) Well answered but some candidates incorrectly referred to connecting to the internet and lost marks.

Q8 **(a)** This part of the question was challenging and some really struggled to answer.

(b) Very well answered.

(c) Many candidates had a good understanding of the term “phishing” and gave good answers.

(d) Once again, this type of question is well received by candidates and well answered.

(e) It was good to see how well this question was attempted as, in the past, it has often been poorly answered.

Q9 Cloud computing was an area that was difficult for some candidates. However, some had been well prepared.

(a) This part of the question was confused with Question 7 Part (d) by some candidates who sought to give networking answers and consequently were not awarded any marks.

(b) Some candidates were able to achieve full marks in this part of the question. However, a significant number were unable to place the advantages in the context of a company.

(c) Generally, well answered. Where candidates lost marks, their answers lacked detail, or they provided two valid features of cloud computing but failed to expand on how these could help an online gamer.

Q10 Candidates should be able to accurately state the names of the laws listed in the specification.

(a) Some failed to get a mark on this question as they were unable to state the name of the law correctly.

(b) Clearly candidates have a good overall knowledge of the Data Protection Act as generally, this part was well answered. Many candidates achieved full marks.

(c) Candidates still do not fully understand the role of the Information Commissioner and the Data Controller. Many attempted to get marks by simply repeating Data Protection laws rather than giving and understanding of the role of each person.

Q11 This was the extended writing question and the quality of written communication question.

(a) Some candidates did not have a real understanding of the impact of digital technology on employment. There is such a wealth of material on this topic. Where candidates did not score well, their answers were not well thought out or planned. Some repeated the same thing without realising that they had already discussed the point.

(b) This question was seeking to get the candidate to think of disadvantages to taking training courses online. It was well attempted, but some candidates incorrectly referred to not getting on the job experience.

Q12 **(a)** This part was well answered, however. many put the word “Batch” in the wrong row and got no marks.

(b) Candidates showed a good understanding of the term ‘backup.’ Many provided detailed answers relating to the topic.

(c) Some candidates mis-interpreted this question and consequently marks were not allocated. These candidates did not provide responses relating to the hard drive or compression.

Assessment Unit 2 Digital Authoring Concepts

Supervising Examiners Overview

This paper contained eight, multi-part, questions. Marks available in various questions parts ranged from one mark to a maximum of eight marks in a matching terms style question. The paper included one question dedicated to the assessment of candidate's quality of written communication, this question was well answered by most entrants with many using full sentences to present their response, with some opting to use bullet pointed lists.

It was evident from candidate responses presented across the papers that a wide range of ability levels were entered for this paper. Candidates did well in the multiple choice and one-mark questions, showing a solid understanding of key concepts. Examiners reported however that the level of language used across the paper varied but that a significant number of candidate responses lacked subject specific terminology, and, in some instances, this prevented candidates from accessing higher level marks in some questions. Overall, the examining team was pleased with overall performance and the main candidate responses indicated that the paper was accessible to all. Most candidates obtained marks in all questions presented across the paper.

The range of questions on the paper allowed for a wide range of candidates' ability to access marks. Some questions proved to be more discriminating whereas there were many that granted candidates of all abilities to access marks.

There appeared to be no evidence to suggest that candidates had insufficient time to complete the paper.

Q1 The multiple-choice questions at the beginning of the paper were well answered by the majority with most candidates being awarded two or three out of a maximum of two marks.

Q2 The focus on this question was on the tools and techniques associated with designing multimedia and database solutions. Some key elements of database theory were incorporated into this question in alignment with the scenario presented as a focus for the question.

(a) & (b) Candidate responses show there was a clear understanding of prototyping and the advantages and disadvantages of its use as a design concept.

(c) This question was well answered showing candidates are aware of and able to identify appropriate uses for the design tools listed in the question elements presented. Some candidates, however, misunderstood the technique required for answering this question and instead of circling the correct term from the options presented in the four sentences, circled several whole sentences from the four options provided.

(d) Overall, this question was answered well by candidates showing a good understanding of interactive features on websites.

(e) to (g) Responses to these question elements indicated most candidates have a clear understanding of the key terms associated with database design and development. A few candidates answered with a repetition of one-to-many/many-to-one answer but overall, most candidates clearly listed two of the three possible answers.

Q3 This question examined candidate understanding of input and output to multimedia and digital applications.

(a) Most candidates were awarded two or three marks out of a possible three in this question, showing a clear understanding of the key features of GUI's. Candidates should, however, be reminded that where more than the requested number of ticks are provided in response to a question, 0 marks will be awarded.

(b) Most candidates were awarded marks for the first part of this question, showing an understanding of the advantages of including high levels of interactivity in computer games, a smaller number were able to correctly provide appropriate responses for the second part to this question which focused on the disadvantages of the same.

(c) Response to this question shows candidates have a detailed knowledge of natural language interfaces however some candidates failed to read Part (iii) correctly and used Tick responses as opposed to TRUE/FALSE as requested in the question stem.

(d) Most candidates were able to correctly identify at least one disadvantage of using a touch screen on a digital interface. Responses were at times replicated with candidates expressing the same disadvantage but in different ways, repeated responses will not be credited twice. Responses referencing the loss of stylus were not awarded marks as the touch screen can still be operated. Candidates should also be reminded of the need to include elaboration in questions where description has been requested in the stem; a significant number of candidates included a correct disadvantage but did not describe how it represented a disadvantage.

Q4 Questions such as this one, which draw upon candidates, controlled assessment, experiences are generally well answered, and this year was no exception.

(a) & (b) Many candidates were correctly able to identify the target audience for the application being designed. They were also able to identify at least two or three ways the app could be made accessible for that target audience.

(c) While most candidates were able to correctly describe methods which can be applied to improve the cross-platform compatibility of the application some candidates focused on the need to test the application rather than on identifying methods that could ensure compatibility before testing. The use of HTML as a development tool was a popular and correct response which was credited by the examining team.

(d) Responses show candidate learning of file types and acronyms has improved from previous years assessment although some candidates incorrectly expanded the acronyms in part (ii) as opposed to identifying a feature of the file formats identified.

Q5 This question focused on multimedia applications and allowed candidates to draw upon their own experiences when presenting responses to various question parts.

(a) Most candidates were able to correctly identify at least one interactive feature provided on websites for online shoppers.

(b) Responses to Part (b) varied. A significant number of candidates focused on bullying as being a disadvantage of live streaming on social media when answering Part (i). This is deemed to be a problem with social media in general and not specific to live-streaming so this response was not accepted by the marking team. In response to Part (ii), being able to socialize with others was a popular but incorrect response, live streaming on social media is a one-way display, others can present likes for the stream, but they are unable to interact with it (at the time of writing). In answering Part (iii), candidates were able to draw upon their own experiences to correctly identify several additional features provided for social media users. In answering questions of this nature in future submissions, candidates should be reminded to take note of the inclusion of the term 'additional' in the question stem. Features referenced elsewhere in the question stem will not be credited.

Q6 This question which referenced multimedia applications from the specification content, included the only QWC question in the paper.

(a) & (b) Both parts were well answered by most candidates. In Part (i), the acronym HTML was correctly expanded by most candidates. A common misconception however was that the 'M' stood for media as opposed to mark-up. Responses to Part (b) showed that candidates have a clear understanding of the functions of the HTML tags included in the specification. Some candidates did, however, confuse the description of the <head> and the <title> tags.

(c) Response to this question varied from zero to three marks with quite a number of candidates being awarded three marks and above. Where candidates' responses were not contextual or incorrect, zero marks were awarded. Candidates who were awarded marks in the level two-mark band were able to correctly name and briefly describe two or three advantages of using authoring applications to create websites. Where candidates were awarded marks in the level three-mark band, they were able to give full elaboration of three advantages and they made good use of technical terminology in their responses.

Q7 Database development questions provide an opportunity for candidates to draw upon controlled assessment learning in an examination context. Response in the main across this question show candidates have a good understanding of the key features of relational database applications and their development. Questions focusing on complex queries and SQL queries, provide more challenge for candidates.

(a) This question was well answered by the majority of candidates showing there is a clear knowledge of the key features of database objects and their uses.

(b) In describing the purpose of a key field in Part (i) of this question, a significant number of candidates responded with generic terms such as 'so it can be unique.' More detail is needed in response to this technical question. Candidates should be encouraged to learn full definitions of this and other key terms. Some candidates had difficulty identifying the foreign key used int eh OrderTBL.

(c) & (d) Most candidates were able to correctly complete the query designs presented in these question parts, showing there is a clear understanding of the steps involved in creating/designing query by example queries in a relational database. Part (d) was a more challenging question which incorporated fields from multiple tables, required candidates to identify the criteria plus determine which fields should be shown as part of the query display. The details provided in the question stem and in the question stimulus meant candidates needed to read all elements of the question carefully. Those who did so were awarded full marks in this question.

(e) Candidates focused on the need to amend the query criteria of Dromore rather than how the address field needed to be changed to ensure the data was correctly structured to support the development of an appropriate query.

(f) Most candidates were correctly able to expand the acronym SQL.

(g) This SQL statement provided an opportunity for candidates of all levels to access marks. Most candidates were able to score at least two out of the four available marks.

(h) Most candidates were able to correctly analyse the features of the report provided in the question stimulus with the majority being awarded at least three out of the available five marks available.

Q8 This question focused on testing techniques and methods and the importance of evaluation in the development process.

(a) Performance in this question was mixed showing that candidates struggle with describing the types of activities performed in the various types of tests carried out during the development of digital solutions. A minority of candidates were awarded full marks in this question with most being awarded partial marks only.

(b) Response to Part (i) showed candidates had a clear understanding that A/B testing involved the use of two versions of the same software, some candidates referenced the use of two applications as opposed to two versions of the same application: preventing them obtaining both of the available marks for this question part. In answering question Part (ii), most candidates were easily able to identify a disadvantage of the use of A/B testing. A large proportion of answers focused on the time/cost element associated with the development of two versions of the same application. Some candidates struggled to identify an appropriate advantage.

(c) This question was well answered by most candidates, those who were awarded one as opposed to two of the available marks, failed to mention the action they would expect to occur after clicking on the navigation buttons during the testing process.

Principal Moderator's Report

Assessment Unit 3 Digital Authoring Practice

Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks electronically. Centres **must** ensure candidates have been provided with the correct scenario for Units 3. This information is clearly provided on the front cover of the controlled assessment units. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised.

Administration

The following procedures **must** be adhered to:

- All candidates of the requested sample, including the **top** and c**omplete bottom candidate** of the sample must be included.
- Only **essential files** should be submitted for moderation. Original data files or duplicate word processing files that have been converted to **Portable Document Format** should not be included. A small number of centres did not meet this particular requirement and submitted duplicate files for each candidate.
- There is no requirement to copy and paste the **mark band information i**nto the eCandidate Record Sheet. Brief comments regarding how marking was allocated are sufficient.

Accuracy of Marking

The accuracy of assessment judgements from centres was generally very high and demonstrated a sound understanding of the controlled assessment requirements. A small number of centres marked leniently, and marking was adjusted. In some cases, marking was slightly harsh at the lower end of the mark range. Centres should use the 'best fit' mark band approach. The use of templates is prohibited. Candidates are encouraged to produce individual solutions. This is a controlled assessment regulation. Assessment of testing and evaluative sections was an area of concern for some centres.

Most centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that all centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures.

It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshots or step by step reports for controlled assessment tasks. Thank you to all centres for their efforts this series with electronic controlled assessment submission.

Task 1 – Designing the multimedia and database solution using appropriate tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- Most storyboards were computer generated. If 'hand-drawn' storyboards are used, they must be scanned to a high resolution to assist the moderation process. Rather than the focus on the quantity of storyboarding, candidates should focus on the quality and detail of storyboards, for example, the design of the website could include the design of the masterpage. Database design may include a more text/tabular form/query/report design, to reduce the duplication of work.
- One example of each query, form, report planning is a requirement. More detailed planning on a masterpage would reduce workload, yet still a produce a high-quality multimedia solution.
- Storyboards must show the sources of information and file types to be used.
- The design of professional reports is to be commended, with full titles and the company logo applied appropriately.
- Some candidates failed to plan complex queries (two or more criteria) or appropriate reports (grouping, sorting, and calculations).
- Some candidates did not attempt prototyping and the use of feedback in refining the solution, this section was marked leniently. 'Marking up' of prototype amendments and feedback could be applied to storyboards in a different font colour.
- Some centres produced unnecessary detail in their design sections e.g. full prototype designs, followed by full storyboards.

Task 2 – Building the multimedia and database solution

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. The following points should be noted:

- Graphics, animation, and videos produced for the candidate websites were generally of a high standard.
- Overall, the standard of web authoring was high. It is positive to note many candidates using masterpages and cascading style sheets.
- Scripted elements may include: form validation, slideshow, animation, login, CSS, colour (hex).
- Accessibility elements may include: browser settings, plugins (downloads), speech, magnification, Alt tags.
- **The location of scripted and accessibility elements must be noted on the eCandidate record sheet.**
- Candidates should be encouraged to name their website homepage as 'index' or 'home.'
- Database solutions incorporated appropriate relationships, user friendly and intuitive switchboards, in addition to appropriate validation and input masks.
- Database solutions were implemented from the design document, however the depth of complex queries and reports (grouping, sorting, and calculations) were lacking, particularly grouping and calculations.

Task 3 – Testing the database solution

- This section was marked leniently by some centres.
- Testing must include a range of valid, invalid, and extreme data.
- Some candidates designed excessive tests and included screenshots for all testing cases, this could be further refined.
- ***No screenshots are required for testing; screenshots are only required to demonstrate corrective action.***

Task 4 – Evaluating the multimedia and database solution

- This section was marked leniently by some centres.
- Many candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met, or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness, and refinements were not always given in depth consideration.

All documentation should be submitted as one PDF.

Chief Examiner's Report

Assessment Unit 4 Digital Development Concepts

Supervising Examiners Overview

The paper contained 11 compulsory questions. There was a range of questions styles including multiple choice, fill the blanks, closed response, tick box, matching terms, and open response questions. Marks allocated to parts of questions ranged from 1 to 8 for completion of algorithmic solutions. In question 8, the candidates' quality of written communication was assessed, this question was worth 6 marks and was marked within levels; there were three levels describing the features of different types of responses from candidates.

The paper covered a wide range of learning outcomes from the specification for this unit.

It was clear from candidate responses that there is a solid understanding of the basic concepts of creating algorithmic solutions and by implication the coding/programming skills required to implement them. A significant number of candidates were able to correctly deduce the outcomes from algorithms presented, identify accurately data types for variables, explain how validation works and perform binary arithmetic.

Questions on arrays or list structures were well answered by some candidates. However, a significant number of candidates had problems with processing an array structure using a loop. In particular these candidates made errors when using the loop counter as the array index. This year, a starting index of 1 was accepted when processing arrays, however, going forward candidates should recognise that arrays are 0 indexed and therefore loops created to process arrays should start at 0.

Responses indicated that candidates were well prepared for the paper. Questions presented allowed for a broad range of candidates to access marks.

Q1 Question one contained some multiple-choice questions and a truth table, for completion by candidates. This was generally very well attempted with a significant number of candidates achieving the full 10 marks available. It was pleasing to see many candidates achieving 6 marks in Part (c) of the question, by completing the truth table accurately.

Q2 Question two was set as a computational thinking exercise. Candidates were asked to consider an algorithm and then to select the correct output given a set of values. Again, this question was well attempted with many candidates achieving 4 marks. Part (c) of this question proved to be the most challenging and some candidates incorrectly selected C as their chosen answer to this part of the question.

Q3 This was a scenario-based question.

(a) Candidates were required to suggest data types for the variables in this scenario. This was generally very well done. Some candidates suggested double for the variable FridaySales. This was not accepted as the scenario was about sandwiches and it would not be possible to have a partial sandwich sale.

(b) **(i)** Candidates were asked to explain the term constant, this was well attempted with many candidates achieving 2 marks. In this response candidates were expected to show an understanding that the constant does not change whilst the programme is running. Some candidates lost marks because they gave vague answers such as "does not change" without expansion.

(ii) This part of the question was less well attempted. Many candidates were unable to articulate a clear advantage for the use of a constant in a programme. Many answers referred to the fact that a constant does not change, which is in fact a response to Part (b)(i) of this question.

(c) This part of the question required candidates to write an algorithm given a detailed set of instructions. Many candidates were able to score highly and achieved full marks in this part of the question. However, some candidates lost marks because their assignment statements were not structured correctly, they did not use the variable names specified in the question or calculations were incorrectly carried out. It is vital that candidates take advantage of the detailed instructions in this type of question and use the variable names supplied. Candidates should ensure that all assignment statements are correctly structured.

Q4 **(a)** This was a fill the blanks question about computational thinking. It was extremely well attempted with a high number of candidates achieving full marks.

(b) Candidates were expected to identify the two symbols presented in the table. The first symbol was a decision box and the second was a process box. A significant majority of candidates correctly identified both boxes and were able to adequately explain how a decision box could be used in a flow chart. The explanation of how the process box symbol could be used was less well attempted. Many candidates giving answers such as 'to show a process.'

(c) **(i) & (ii)** It was clear from responses in these parts of the question that candidates had a good understanding of syntax errors, logical errors, and runtime errors. These parts of the questions were well attempted, and many candidates achieved full marks.

(iii) This part of the question was more challenging for many candidates. A significant number of candidates achieved 2 marks out of a possible 4. In these cases, candidates were able to state an advantage but were unable to expand it. For example, whilst code re-usability is an advantage, candidates should be able to expand on the impact of the advantage by referring to, for example, shorter development time.

(d) Candidates were required to write a validation loop for a variable, option. The instructions in this part of the question were detailed so that, candidates would be able to access all the 8 marks available. This worked for a significant number of candidates who achieved full marks in this part of the question. Some candidates lost marks as they used a FOR loop which is unsuitable for this application. Other reasons for losing marks included incorrect conditions set in the loop, incorrect exit logic and conditions which were not constructed correctly (for example inequalities in the wrong order).

The use of While True and Break where acceptable within the loop structure.

Q5 **(a) (i) (ii) &(iii)** Parts (i) and (ii) of this question relating to a length check and a presence check were well attempted by most candidates. The format check proved a little more challenging and a number of candidates lost marks because they did not refer to the special symbols or capital letter specified in the scenario. The question specifically asked how the validation checks could be used when validating Amelia's new password therefore, candidates should have referred to the examples within the question.

(b) Part (b) of this question was less well answered. With four marks available, candidates were expected to identify a string function and explain how it could help when validating the password. The specification states that candidates should be able to demonstrate an understanding of the use and the functionality of string manipulation within a programming language. This includes string splitting, concatenating, character searching and substring searching. A significant number of candidates were unable to name a string function related to any of these.

Q6 This was another scenario-based question. When attempting questions like this, candidates should be encouraged to read carefully the information provided as it is structured to help them complete or create algorithms.

(a) Candidates were required to complete an algorithm and this part of the question was very well attempted by most candidates with a significant proportion of candidates achieving full marks.

(b) This part of the question required candidates to think about how data is stored when a programme is not running. The expected response was to use a file. However, some candidates incorrectly referred to data structures such as arrays. Many candidates however, correctly referred to CSV files and text files and were credited with full marks.

Q7 This question was about binary arithmetic. Throughout the question, candidates demonstrated an excellent knowledge and understanding of converting numbers from the various different bases. A significant majority of candidates successfully added the two bytes together and achieved 3 marks in the calculation. Descriptions of how overflow affects the result of a calculation were very good. Overall, this question was very well answered, and a broad range of candidates were able to access the marks available.

Q8 The quality of written communication was assessed in this question. Candidates were required to explain how a software development environment helps a programmer when creating and running code. Candidates were prompted to include editing features and high-level code translation and execution, in their answer. There were some excellent answers to this question attracting the full 6 marks. Candidates falling into the middle band typically described only one of the features or provided are very weak explanation of one of the features. Those candidates who were assigned Level 1 failed to expand adequately on the features and provided very little in the way of examples.

Q9 This question, in its entirety, focused on arrays or list structures. Candidates were presented with a table of data representing the data stored in the array.

(a) **(i)** This part of the question was well attempted by candidates. The majority selected the correct answer which was integer. Candidates selecting double or real were not given credit as the figures presented in the table were whole numbers. Candidates should be encouraged to study the data provided and to take note of the data type presented in the question.

(ii) Many candidates provided the correct assignment statement to change the value of the 5th element in the array. This year pupilHeights[5] was accepted. However, as outlined in the overview of the examination, going forward students will be expected to know that an array or list structure is 0 indexed.

(iii) In this part of the question candidates were required to write an algorithm, using a loop that set all the values in the array to 0. Again, candidates found difficulty with using the loop counter as the array index. Some candidates were unable to formulate the loop correctly. Many candidates achieved 4 marks by creating a simple FOR loop, running from 0 to 9 (or 0 to array.Length) and assigning the correct element in the array to 0. FOREACH was accepted.

(b) Candidates were required to complete an algorithm which searched through an array to find the maximum value. Whilst a significant number of candidates achieved full marks in this question, many candidates were unable to accurately state the array index and the value that maxheight would be set to inside the IF statement. This led to many candidates achieving 2 marks out of a possible 4. The two marks were awarded for inserting maxheight into the if statement and outputting maxheight at the end of the algorithm. Candidates clearly understood the logic of the problem. However, some candidates have difficulty in writing down the correct syntax when processing an array. Candidates need to be aware that the array name followed by the index is required when referring to a single element within an array.

(c) **(i)** This part of the question, related to the bubble sort, was a stretch and challenge question. Most candidates were able to identify the second statement in the sequence and a significant number identified the correct sequence for the following four statements. Again, this question required an understanding of array processing. It was interesting to note that candidates found ordering the last four statements the most difficult. Candidates are expected to have an understanding of how the bubble sort works and the order of statements in a simple representation of it.

(ii) Correctly answered by a significant number of candidates.

(iii) Most candidates were able to provide a search technique however, some chose the linear search which was incorrect.

Q10 **(a)** This part of the question which involved completing a paragraph about testing, was well attempted by most candidates. Many candidates scored a full 5 marks. Common mistakes included inserting unit instead of system or system instead of integration.

(b) Part (b) of the question focused on different types of test data that is: valid, invalid, and extreme. Candidates were required to explain the different types of test data and provide examples from the scenario given in the question. The part pertaining to invalid data was well attempted by most candidates, many providing a correct example of invalid data. Extreme data was less well attempted and candidates did not refer to upper and lower boundaries or maximum and minimum values. Many did not provide a valid example of extreme data which in this case was 1 or 6.

Q11 The final question in the paper was on the evaluation process.

(a) Candidates were required to describe how the user requirements and design documents could be used when evaluating an application. Some candidates achieved full marks in this question, but a significant number lost one or two marks because they correctly identified the function of these two documents but failed to provide an adequate explanation of how the documents could help the developer when evaluating the application. Candidates should be encouraged to read questions carefully so that they include all aspects of the required response.

(b) Well attempted by most candidates illustrating that candidates have a good understanding of the fact that evaluation is a continuous process, can help ensure that user requirements are met and can be used to help get feedback from the end user.

Principal Moderator's Report

Assessment Unit 5 Digital Development Practice

Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks electronically. Centres **must** ensure candidates have been provided with the correct scenario for Unit 5. This information is clearly provided on the front cover of the controlled assessment units. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised.

Administration

The following procedures **must** be adhered to:

- All candidates of the requested sample, including the **top** and co**mplete bottom candidate** of the sample must be included.
- Only **essential files** should be submitted for moderation. Original data files that have been converted to **Portable Document Format** should not be included. A very small number of centres did not meet this particular requirement.

Accuracy of Marking

The accuracy of assessment judgements from centres was generally very high and demonstrated a sound understanding of the controlled assessment requirements. High quality work was produced by centres and application of the mark scheme was highly accurate.

A small number of centres marked leniently, and marking was adjusted. The use of templates is prohibited. Candidates are encouraged to produce individual solutions. Assessment of testing and evaluative sections continue to be an area of concern for some centres, regarding leniency.

Most centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that **all** centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures.

It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshots for controlled assessment tasks. Thank you to all centres for their efforts this series with electronic controlled assessment submission.

Task 1 – Design a Solution using Appropriate Tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- In some instances, user requirements were extracted from the task, without development or interpretation. Target audience needs must also be fully identified.
- The design component was particularly well implemented with high quality flowcharts and pseudo code included effectively.
- Validation and error trapping techniques were designed appropriately, however, use of user feedback to refine the solution was omitted by a number of centres and marked leniently.

Task 2 – Building a Solution

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. Centres are to be highly commended upon their approach to this task. The following points should be noted:

- **Program usernames and passwords must be noted on the eCandidate record sheets. A number of centres are not adhering to this essential element of controlled assessment submission.**
- Marking was highly accurate across the sampled centres and the programming solutions were of a high standard.
- Candidates should be encouraged to develop their own program style and content.
- Solutions of the highest standard gave consideration to system robustness.
- All solutions must be annotated.

Task 3 – Testing the Solution

- Generally, test plans were well structured, in tabular format (showing tests, expected outcome, actual outcome and fixes), in some cases test plans were cluttered and could be more professionally presented.
- It is very important that testing incorporates a range of tests (black and white box testing) and includes valid, invalid, and extreme test data. Extreme testing of data was lacking in a small number of centres.
- Some candidates designed excessive tests and included screenshots for all testing cases.
- ***No screenshots are required for testing; screenshots are required to demonstrate corrective action.***

Task 4 – Evaluating the Solution

- Some candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met, or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness, and refinements were not always given in depth consideration.
- A small number of centres marked performance, robustness, and refinements section leniently.

All documentation should be submitted as one PDF.

Contact details

The following information provides contact details for key staff members:

- **Specification Support Officer: Nuala Tierney**
  (telephone: (028) 9026 1200, extension: 2292, email: ntierney@ccea.org.uk)
- **Officer with Subject Responsibility: Andrew Douglas**
  (telephone: (028) 9026 1200, extension: 2713, email: adouglas@ccea.org.uk)

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COUNCIL FOR THE CURRICULUM, EXAMINATIONS AND ASSESSMENT
29 Clarendon Road, Clarendon Dock, Belfast BT1 3BG
Tel: +44(0)28 9026 1200 Email: info@ccea.org.uk
Web: ccea.org.uk

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